

Product Requirement Document (PRD)

Smart ETA Prediction & Delivery Optimization System

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Industry:

Food Delivery & Marketplace Operations

1. Executive Summary

This document proposes a Smart ETA Prediction & Delivery Optimization System for a food delivery marketplace platform.

The feature aims to improve delivery time accuracy using:

- *Real-time rider availability*
- *Traffic conditions*
- *Restaurant preparation time*
- *Historical delivery data*
- *Peak-hour demand patterns*

The primary objective is to reduce order cancellations caused by inaccurate delivery estimates while improving customer trust, operational efficiency, and repeat order rates.

2. Business Problem

Customers frequently experience delays between estimated delivery time and actual order completion.

Inaccurate ETAs negatively impact:

- Customer trust
- User retention
- Repeat orders
- Marketplace reliability

Current ETA systems rely on static assumptions and fail to dynamically adjust for:

- Traffic congestion
- Rider shortages

- Restaurant preparation delays
- Peak-hour order spikes
- Weather conditions

This results in:

- Increased cancellations
 - Higher support complaints
 - Lower customer satisfaction
 - Reduced operational efficiency
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3. Business Impact

The current delivery experience directly impacts customer retention and operational efficiency across the marketplace ecosystem.

Inaccurate ETAs contribute to:

- Increased order cancellation rates
- Higher customer support volume
- Reduced customer trust
- Lower repeat order frequency
- Negative app ratings and reviews

Operationally, delayed deliveries also reduce rider utilization efficiency and create delivery bottlenecks during peak demand periods.

Improving ETA accuracy is expected to:

- Reduce cancellation rates
 - Increase repeat orders
 - Improve customer satisfaction
 - Optimize rider allocation
 - Enhance marketplace reliability
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4. Product Goals

The Smart ETA Prediction & Delivery Optimization System aims to improve delivery transparency and operational efficiency across the marketplace platform.

Primary product goals include:

- Improve ETA prediction accuracy
 - Reduce order cancellation rates
 - Increase repeat customer orders
 - Improve customer trust in delivery timelines
 - Optimize rider allocation efficiency
 - Reduce support tickets related to delayed deliveries
 - Improve overall marketplace reliability during peak demand periods
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5. Success Metrics (KPIs)

The success of the Smart ETA Prediction & Delivery Optimization System will be measured using operational, customer experience, and marketplace performance metrics.

KPI	Business Impact
ETA Accuracy %	Improves customer trust
Order Cancellation Rate	Measures failed deliveries
Repeat Order Rate	Measures customer retention
Average Delivery Time	Operational efficiency metric
Customer Satisfaction (CSAT)	User experience quality
Rider Utilization Rate	Delivery efficiency
Support Ticket Volume	Operational issue tracking

6. User Personas

The Smart ETA Prediction & Delivery Optimization System impacts multiple stakeholders across the marketplace ecosystem.

1. Customer Persona

Profile:

Users placing food delivery orders through the platform.

Goals:

- Receive accurate delivery estimates
- Track orders reliably
- Avoid unexpected delays

Pain Points:

- Delivery arriving later than promised
- Lack of trust in ETA timings
- Poor communication during delays

2. Rider Persona

Profile:

Delivery riders responsible for order fulfillment.

Goals:

- Efficient delivery routing
- Balanced order allocation
- Reduced waiting times at restaurants

Pain Points:

- Traffic congestion
- Restaurant preparation delays
- High delivery pressure during peak hours

3. Operations Team Persona

Profile:

Internal operations and marketplace monitoring teams.

Goals:

- Reduce delivery delays
- Optimize rider allocation
- Improve operational efficiency

Pain Points:

- High cancellation rates
 - Peak-hour delivery bottlenecks
 - Increased support escalations
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7. Functional Requirements

The Smart ETA Prediction & Delivery Optimization System will dynamically calculate and continuously update estimated delivery times throughout the customer journey.

Requirement ID	Feature Requirement	Priority
FR-01	System should calculate ETA using live traffic data	High
FR-02	System should factor restaurant preparation time into ETA	High
FR-03	System should dynamically update ETA during delivery	High
FR-04	System should notify users when delivery delays occur	Medium
FR-05	System should optimize rider allocation during peak hours	High
FR-06	System should provide ETA confidence scoring internally	Medium

8. User Stories

The following user stories define expected user interactions and feature outcomes for key stakeholders within the marketplace ecosystem.

User Type	User Story	Business Value
Customer	As a customer, I want accurate delivery estimates so that I can plan my time effectively	Improves customer trust
Customer	As a customer, I want delay notifications so that I remain informed during delivery delays	Reduces frustration
Rider	As a rider, I want optimized order allocation so that I can complete deliveries efficiently	Improves rider utilization
Operations Team	As an operations manager, I want real-time ETA monitoring so that I can proactively manage delivery delays	Improves operational visibility
Support Team	As a support agent, I want access to ETA prediction data so that I can better handle customer complaints	Improves support efficiency

9. Analytics & Event Tracking Plan

The platform will track key user interactions and operational events to measure delivery performance, customer experience, and feature effectiveness.

Event Name	Purpose	KPI Impact
ETA_Viewed	Tracks customer ETA visibility	Customer engagement
ETA_Updated	Tracks dynamic ETA changes	ETA accuracy
Order_Cancelled	Measures cancellation reasons	Cancellation rate
Delivery_Delayed	Tracks delivery delays	Operational efficiency
Rider_Assigned	Measures rider allocation timing	Rider utilization
Support_Ticket_Created	Tracks customer complaints	Customer satisfaction

10. Risks & Assumptions

The implementation and effectiveness of the Smart ETA Prediction & Delivery Optimization System depends on operational consistency, data quality, and marketplace conditions.

Risk / Assumption	Potential Impact
Inaccurate traffic data	Incorrect ETA predictions
Rider shortages during peak hours	Increased delivery delays
Restaurant preparation inconsistencies	ETA deviations
GPS tracking inaccuracies	Poor delivery visibility
Unexpected weather conditions	Delivery slowdowns

11. Rollout Strategy

The Smart ETA Prediction & Delivery Optimization System will be deployed in phased rollout stages to minimize operational disruption and validate performance improvements incrementally.

Phase	Rollout Scope	Objective
Phase 1	Internal testing environment	Validate ETA prediction logic
Phase 2	Limited city rollout	Monitor operational performance
Phase 3	Peak-hour deployment testing	Evaluate high-demand behavior
Phase 4	Full marketplace rollout	Scale feature across platform

12. Conclusion & Expected Outcomes

The Smart ETA Prediction & Delivery Optimization System is designed to improve delivery transparency, operational efficiency, and marketplace reliability within food delivery platforms.

By leveraging dynamic ETA prediction capabilities, the platform is expected to:

- Improve customer trust in delivery timelines
- Reduce order cancellation rates
- Increase repeat customer orders
- Improve rider allocation efficiency
- Enhance operational visibility during peak demand periods

The proposed solution aligns with marketplace growth objectives by improving both customer experience and delivery operations at scale.
